

NeuralRetail

AI Sales Intelligence Platform

Internship Project • **Amdox Technologies**

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Data Science & Analytics

541K+

Transactions

38

Countries

5

Dashboard Pages

3

ML Models

Project Overview

NeuralRetail is an end-to-end AI-powered sales intelligence platform built on the UCI Online Retail Dataset — 541,909 real transactions across 38 countries. The platform combines machine learning, customer analytics, and interactive dashboards to deliver actionable business intelligence.

📄 Churn Prediction

Identify customers at risk of leaving using XGBoost & LightGBM with AUC-ROC ≥ 0.90

📄 Demand Forecasting

30-day revenue forecast with Linear Regression + lag features targeting MAPE $\leq 10\%$

📄 Customer Intelligence

RFM-based segmentation into Champions, Loyal, At Risk, and Hibernating clusters

📄 Model Tracking

MLflow experiment tracking with Evidently AI drift detection for production monitoring

Technology Stack

Language

Python 3.12

ML Models

XGBoost • LightGBM • Scikit-learn

Explainability

SHAP (SHapley Additive Explanations)

Experiment Tracking

MLflow

Drift Detection

Evidently AI

Dashboard

Streamlit + Plotly

REST API

FastAPI + Uvicorn

Data Processing

Pandas • NumPy

Dataset & Exploratory Data Analysis

541,909

Total Transactions

4,338

Unique Customers

3,665

Unique Products

38

Countries

£8.88M

Total Revenue

£479

Avg Order Value

Key EDA Findings

- UK dominates revenue (~84%) followed by Netherlands, Germany & France
- November–December peak season — highest revenue months across all years
- 33.3% customer churn rate identified — 1,445 churned vs 2,893 active customers

Customer Intelligence — RFM Segmentation

R

Recency

How recently did the customer purchase?

F

Frequency

How often do they purchase?

M

Monetary

How much do they spend?

Customer Segments Identified

Champions

27.2% of customers

Bought recently, buy often, spend most

Loyal Customers

14.9% of customers

Regular buyers with strong spending

At Risk

15.6% of customers

Used to buy often — now slipping away

Hibernating

20.3% of customers

Long inactive, low spend

Lost

14.0% of customers

Lowest recency, frequency & spend

Potential Loyalists

8.0% of customers

Recent buyers with growth potential

Churn Risk Prediction

33.3%

Overall Churn Rate

1,445

Churned Customers

2,893

Active Customers

AUC $\geq$ 0.90

Model Target

XGBoost Churn Model

- Algorithm: Gradient Boosted Trees
- Metric: AUC-ROC (target $\geq$ 0.90)
- Hyperparams: `n_estimators`, `max_depth`, `learning_rate`
- Tracking: MLflow experiment logging

LightGBM Churn Model

- Algorithm: Light Gradient Boosting Machine
- Metric: AUC-ROC (target $\geq$ 0.90)
- Faster training, handles large feature sets
- Compared against XGBoost for best performance

SHAP Explainability: Feature importance analysis performed to identify top churn drivers — Recency, Frequency, and Monetary scores were the dominant predictors.

Demand Forecasting

Forecasting Approach

01 Feature Engineering

Lag features (7, 14, 30-day), rolling means, day-of-week encoding from daily sales data

02 Model Training

Linear Regression trained on engineered features with 30-day forecast horizon

03 Evaluation

MAPE metric used — target $\leq 10\%$. Actual vs Predicted plotted interactively in dashboard

04 MLflow Tracking

All runs logged with parameters, metrics, and model artifacts for reproducibility

Result: MAPE achieved was 23.2% — above the 10% target, which is expected for a linear model on retail data with seasonal spikes. This is noted as a limitation and improvement opportunity.

Dashboard & REST API

5-Page Streamlit Dashboard

📄 Executive Overview

Revenue KPIs, monthly trends, country breakdown, top 10 products

📄 Demand Forecast

Actual vs predicted revenue chart, MAPE metric, historical daily sales

📄 Customer Intelligence

RFM segment pie chart, revenue by segment, distribution histograms

📄 Churn Risk Analysis

Churn distribution, segment risk, SHAP chart, high-risk customer table

📄 Model Tracker

MLflow experiment summary, live model file status, evidently reports

FastAPI REST API

Framework: FastAPI + Uvicorn
Auto docs: /docs (Swagger UI)

Endpoints:

POST /predict/churn
GET /forecast
GET /segments
GET /health

Accepts customer RFM
features as JSON input
Returns churn probability
and segment classification

MLflow Tracking & Drift Detection

MLflow Experiment Tracking

Models Tracked

XGBoost, LightGBM, Linear Regression

Params Logged

n_estimators, max_depth, learning_rate, subsample

Metrics Logged

AUC-ROC, Accuracy (churn) | MAPE (forecast)

Artifacts Saved

model.pkl / model.xgb files per run

Run History

Multiple experiment runs stored locally

Evidently AI — Drift Detection

Data Drift Report

Detects feature distribution shifts in production data

Missing Values

Tracks missing value patterns across pipeline runs

Confusion Matrix

Visual performance breakdown for churn model

SHAP Importance

Feature-level explainability chart generated

Output Format

HTML reports saved to evidently_reports/ folder

Command to view MLflow UI locally: `mlflow ui --backend-store-uri ../mlflow_runs` → `http://localhost:5000`

Project Structure & Deliverables

Folder Structure

```
NeuralRetail/  
├── data/           ← Datasets & CSVs  
├── notebooks/      ← 4 Jupyter Notebooks  
├── src/  
│   ├── api/        ← FastAPI endpoints  
│   └── models/      ← Trained .pkl files  
├── dashboard/  
│   └── app.py       ← Streamlit (5 pages)  
├── evidently_reports/ ← Drift HTML reports  
├── mlflow_runs/     ← Experiment history  
├── assets/screenshots/ ← Dashboard images  
└── README.md
```

Deliverables Completed

- EDA Notebook (01_EDA.ipynb)
- Feature Engineering (02_feature_engineering.ipynb)
- Model Training (03_model_training.ipynb)
- Drift Detection (04_drift_detection.ipynb)
- 5-Page Streamlit Dashboard
- FastAPI REST API with Swagger docs
- MLflow Experiment Tracking
- Evidently AI Drift Reports
- Live Deployment on Streamlit Cloud
- GitHub Repository (public)

Conclusion & Key Learnings

🔗 Built a complete ML pipeline — from raw data to deployed production dashboard

🔗 Gained hands-on experience with MLflow tracking and Evidently AI drift detection

🔗 🔗 Understood the gap between model metrics and real-world business targets (MAPE)

🔗 Deployed a live Streamlit app with FastAPI backend accessible via public URL

🔗 Applied RFM segmentation to drive real customer intelligence insights

🔗 Live Demo: neuralretail-amdox.streamlit.app 🔗 GitHub: github.com/shubhamjais04/NeuralRetail

Thank You

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